AcuBQA: A RAG-based Bilingual Acupuncture QA System via Domain-Specific Fine-tuning

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***Abstract*—Large language models (LLMs) are prone to hallucination and often lack structured and domain-specific knowledge in Traditional Chinese Medicine (TCM). We present AcuBQA, the first trustworthy bilingual (Chinese–English) acupuncture question-answering system built on fine-tuned LLMs with retrieval-augmented generation (RAG). The system integrates a curated ontology of 361 acupoints and 14 meridians, clinician-authored cases, and a triple-constraint framework (i.e., terminology checking, evidence grounding, and safety filtering) achieved by Direct Preference Optimization (DPO) to produce controlled and verifiable answers. On our evaluation suite, the best configuration (BERT+Baichuan) achieves 94.4% context recall, 97.2% faithfulness, and 96.1% relevance (RAGAS), alongside 0.88 BLEU and 0.94 ROUGE scores. It also received high professional ratings with a 94/100 score from GPT-4 and a 90/100 score from acupuncture experts). After Fine-tuning, the safety metrics including the Safe Refusal Rate (SSR) improved by approximately 20% while the Harmful Generation Rate (HGR) and Jailbreak Success Rate (JBR) significantly decreased. These results show that bilingual embedding fusion and constraint-based Fine-tuning markedly improve factuality, safety and clinical utility over vanilla LLM baselines, providing a strong foundation for reliable and accessible acupuncture-specific QA.**

***Index Terms*—RAG, Question Answering, Chinese Acupuncture, LLMs, Traditional Chinese Medicine, Reinforcement Learning from Human Feedback**

I. Introduction

Large language models (LLMs) have become a central approach in natural language processing, showing strong performance across diverse tasks. Their adoption in safety-critical areas such as healthcare, however, remains constrained by two issues: the tendency to produce hallucinations and the lack of access to structured, domain-specific knowledge. Retrieval-Augmented Generation (RAG) has emerged as a promising paradigm to mitigate these issues by grounding model responses in external knowledge [1]–[3]. In medical QA, text-based RAG systems have demonstrated improved factuality and reduced hallucination rates, making them well suited for clinical decision support and education [1]–[5].

Direct Preference Optimization (DPO) introduces a reparameterization of the reward model in RLHF that enables extraction of the corresponding optimal policy in closed form, allowing us to solve the standard RLHF problem with only a simple classification loss [6]. Recent work has demonstrated that DPO can effectively refine healthcare LLMs against domain-specific safety signals, achieving up to 42% improvement in harmful query detection [7]. DPO alignment has also been adopted in open medical LLMs to emphasize ethical and policy-aligned performance in the presence of jail breaking attacks [8].

Despite these advances, most RAG applications in medicine focus on Western biomedical corpora, such as diagnostic reasoning or radiology reporting [1]–[3]. Traditional Chinese Medicine (TCM) has received much less attention, and within this domain, ***acupuncture***—a globally recognized and clinically important branch—remains almost entirely unexplored from a retrieval-grounded perspective. Acupuncture knowledge involves structured mappings between acupoints, meridians, syndrome differentiation, and treatment strategies, which general-purpose LLMs often fail to capture reliably. Moreover, no prior system has attempted to support bilingual access, despite the growing international demand for acupuncture education and clinical reference [9], [10].

To address these gaps, we present a bilingual acupuncture QA system that integrates a curated ontology of acupoints and meridians with LLM backbones. The system employs a pipeline featuring retrieval-augmented generation (RAG), cross-lingual alignment, and safety post-training which are all grounded in clinical guidelines and patient records. Unlike prior work that emphasizes large-scale biomedical datasets or multi-model reasoning [1]–[3], our system focuses specifically on acupuncture, providing structured, ontology-grounded answers in both Chinese and English. In addition, we incorporate clinician-authored case records to enhance personalization, ensuring that outputs are both context-aware and clinically

relevant. Our contributions are as follows:

- **First Acupuncture-Specific Knowledge Framework:** We address a critical gap in TCM digitalization by developing the first bilingual QA system specifically focused on acupuncture. Unlike existing TCM systems centered on herbal medicine, our system integrates a high-precision ontology covering 361 acupoints and 14 meridians, including specialized clinical parameters such as needling angles, insertion depths, disease-symptom mappings, and therapeutic techniques.
- **Synergistic RAG-DPO Optimization:** We propose a retrieval-augmented generation pipeline enhanced with Direct Preference Optimization (DPO) to provide the trustworthy acupuncture QA. We use RAG to provide the factual grounding via structured JSON knowledge and clinical cases, while DPO is utilized to further align the LLM with expert-level clinical reasoning and safety constraints to effectively reduce hallucinations and unsafe recommendations in acupuncture-related responses.
- **Bilingual Cross-Lingual Accessibility:** We implement a robust cross-lingual alignment strategy that enables the system to process queries and retrieve knowledge seamlessly across English and Chinese. This facilitates the international dissemination of standardized acupuncture knowledge and supports global practitioners with verified and evidence-based information.
- **Empirical Evaluation and Expert Validation:** We conduct an extensive evaluation suite including ablation studies to verify the necessity of each component and benchmarking against multiple SOTA LLMs. The system is further validated through blind reviews by professional TCM experts, achieving a 94/100 professional rating to support its clinical reliability.

II. Related Work

A. RAG for Bilingual Medical QA

RAG has been widely applied in biomedical QA to enhance factuality by grounding model outputs in structured knowledge. Systems such as MMed-RAG [1] and MedRAG [2] demonstrate improved accuracy in medical reasoning via domain-aware retrieval and knowledge-graph–elicited reasoning, while Visual RAG [3] explores retrieval grounding for clinical reporting tasks. Structured retrieval and end-to-end optimization further improve downstream reliability and efficiency beyond fixed retrievers [4], [5].

In contrast, most existing TCM QA systems are monolingual (Chinese-only), limiting accessibility for international learners and practitioners [11]–[13]. Concurrently, acupuncture is increasingly taught and practiced worldwide, creating a concrete need for bilingual educational and clinical tools. By combining retrieval grounding with bilingual query/answer support and Chinese-English alignment, our system fills this gap and demonstrates the feasibility of lightweight, bilingual RAG in the acupuncture domain [9].

B. LLM-Based QA in TCM and Acupuncture

Recent TCM-oriented work has introduced domain-pretrained LLMs, knowledge-graph–augmented QA, and RAG-inspired assistants for education and clinical support (e.g., TCMLCM, medical-education RAG, and TCM knowledge-graph QA) [11]–[13]. More recent systems explore GraphRAG for TCM diagnosis (OpenTCM) [14], tree-organized self-reflective retrieval (TOSRR) [15], multi-round medical RAG (MRDRAG) [16], as well as a standardized TCM QA dataset (TCMD) [17] and applied RAG QA robots [18]. However, these efforts remain monolingual in most cases and target broad TCM knowledge rather than acupuncture-specific ontology with retrieval grounding; to our knowledge, no RAG-based bilingual acupuncture QA has been reported [11]–[18]. Large-scale evaluations of LLMs in TCM further highlight reliability gaps that motivate retrieval grounding and domain structuring [9].

C. Hallucination Mitigation

Approaches to reduce hallucinations include RLHF, domain-specific Fine-tuning, and prompt engineering; Retrieval augmentation provides a lightweight, controllable alternative by grounding outputs in curated evidence [1]–[4]. In healthcare RAG, explicit structure (e.g., knowledge graphs/ ontology) improves factuality and decision specificity [2], [11], [14], [15]. Our system follows this direction by leveraging an acupuncture ontology (acupoints–meridians–syndrome–operation relations) as the backbone for retrieval and reasoning, then combined it with domain-specific Fine-tuning to improve its safety.

III. Methods

A. Overall Framework

Our system adopts a lightweight LLM combined with a Retrieval-Augmented Generation (RAG) architecture [1]–[3] and a a Fine-tuning structure, as shown in Figure 1. Rather than relying solely on large-scale general-purpose LLMs, we ground model outputs in domain-specific acupuncture knowledge to ensure factual accuracy, controllability, and bilingual accessibility. The framework is organized into four layers: a knowledge base construction layer that encodes structured acupuncture knowledge, a retrieval-augmented generation pipeline that integrates embeddings, retrieval, and generation, a Fine-tuning layer that ensures bilingual alignment and secure choice, and an application layer that delivers bilingual answers via a web-based front end. The overall process can be formalized as

$$y = \mathrm{LLM}(q, \{d_1, d_2, \ldots, d_k\}), \tag{1}$$

where q is the user query, $\{d_i\}$ are the retrieved passages, and y is the final constrained response.

B. Knowledge Base Construction and Response

The acupuncture knowledge base is key to system reliability. As shown in Figure 2 and Figure 3, we first digitize classical TCM texts and clinical manuals via OCR and clean the data to

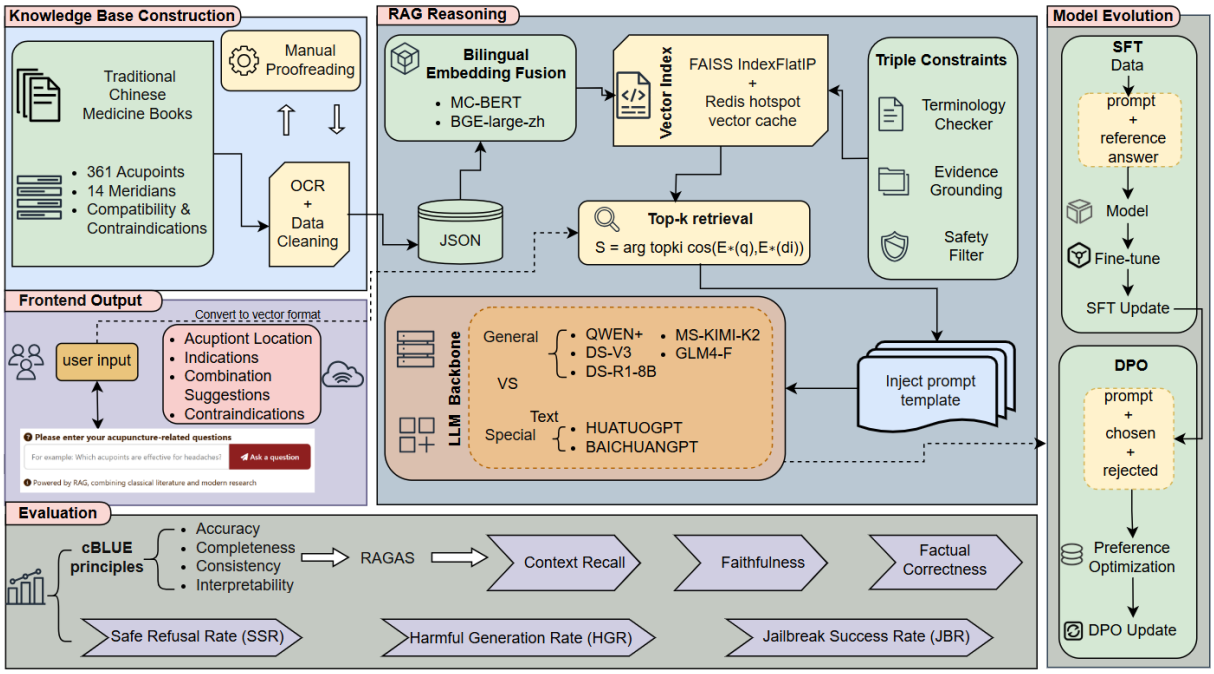


Fig. 1: System framework of the bilingual acupuncture RAG QA system. It integrates a knowledge base, RAG reasoning pipeline, and application frontend.

remove noise. We then define an ontology-based schema with entities (acupoints, meridians, syndromes, contraindications) and relations (belongs_to, indicated_for, contraindicated_for). The data are organized into structured JSON files, where each acupoint includes location, indications, operations, and pairing rules. To support bilingual retrieval, we adopt a parallel embedding paradigm with two language-specialized encoders. We first extract monolingual embeddings separately: Chinese text is encoded via BGE-large-zh [19] and English text is processed by MC-BERT [20], yielding two isolated language-specific embedding spaces. Afterwards, we attach lightweight projection heads to each encoder, and optimize cross-lingual contrastive loss on parallel bilingual sentence pairs to align the two heterogeneous spaces into a unified cross-lingual vector space, without modifying the pre-trained parameters of the two encoders. These vectors are indexed in FAISS using cosine similarity,

$$E(q) = \text{Encoder}(q), \tag{2}$$
$$S(q, d_i) = \cos(E(q), E(d_i)), \tag{3}$$

allowing efficient top-k retrieval. Retrieved knowledge passages are then injected into the model prompt template to produce domain-grounded responses that are both bilingual and ontology-compliant.

C. Clinical reasoning order

To ensure clinical coherence and auditability, the system follows a fixed reasoning sequence aligned with acupuncture practice and our ontology: it identifies the *chief complaint* and *disease/syndrome entities*; performs *TCM syndrome differentiation* (including deficiency–excess patterns and an auxiliary step for complex cases); formulates the *acupoint prescription* by selecting *primary* and *adjunct* points; and finally specifies the *operation* (needling method, depth, manipulation) and lists *contraindications*. In Chinese: "First identify the main complaint and disease, then classify TCM syndromes (with intelligent analysis), choose primary and secondary points, provide needling techniques, and state contraindications." This sequence is enforced by the prompt template and the triple-constraint decoder to ensure evidence grounding and ontology-aligned terminology.

D. Question Generation and Fine-tuning

To strengthen robustness, better capture acupuncture reasoning, and enhance safety alignment, the knowledge base is augmented with synthetic QA pairs derived from clinician cases and classical texts. These pairs support instruction tuning, aligning the model with domain-specific answer styles. Fine-tuning employs a multi-objective loss combining cross-entropy for fluency, a contrastive loss to improve retrieval relevance, and DPO to enforce safety and clinical appropriateness:

$$\mathcal{L} = \mathcal{L}_{CE}(y, \hat{y}) + \lambda \cdot \mathcal{L}_{contrast}\big(E(q), E(d^+), E(d^-)\big) + \gamma \cdot \mathcal{L}_{DPO}(\pi_\theta, \pi_{ref}, (x, y_w, y_l)) \tag{4}$$

where d^+ and d^- denote positive and negative passages, respectively. The cross-entropy term $\mathcal{L}_{CE}$ ensures fluency and domain coherence, while $\mathcal{L}_{contrast}$ anchors responses to clinically valid evidence. The hyperparameter λ balances these two objectives.

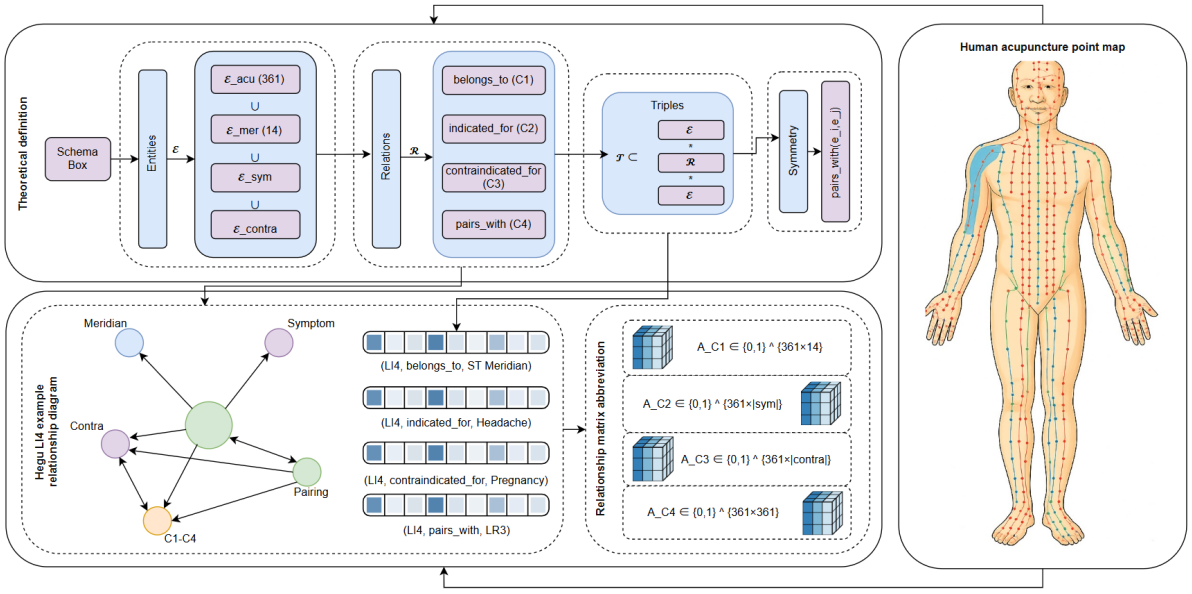


Fig. 2: Acupuncture knowledge graph covering 361 acupoints, 14 meridians, symptoms, and contraindications.

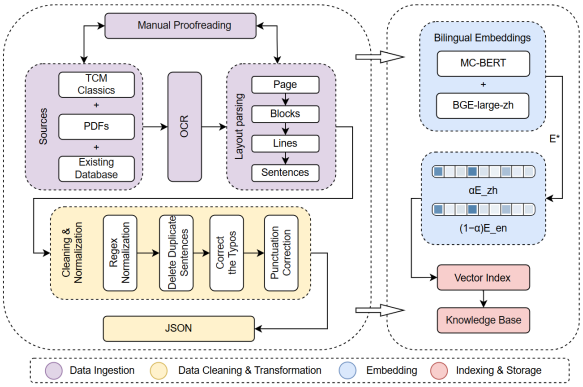


Fig. 3: Knowledge construction pipeline: OCR → cleaning → bilingual embedding → FAISS indexing.

To further enforce safety and clinical appropriateness, a direct preference optimization (DPO) term $\mathcal{L}_{DPO}$ is incorporated. Given a prompt x, a preferred (safe/clinically appropriate) response y_w, and a dispreferred (unsafe/inappropriate) response y_l, the DPO objective is formulated as:

$$\mathcal{L}_{DPO}(\pi_\theta, \pi_{ref}, (x, y_w, y_l)) = -\log \sigma \left(\beta \log \frac{\pi_\theta(y_w|x)}{\pi_{ref}(y_w|x)} - \beta \log \frac{\pi_\theta(y_l|x)}{\pi_{ref}(y_l|x)} \right) \quad (5)$$

where π_θ denotes the fine-tuned policy model, π_{ref} is the fixed reference model, β controls the strength of preference alignment, and $\sigma(\cdot)$ is the sigmoid function. The hyperparameter γ governs the relative weight of safety alignment in the overall training objective. This multi-objective formulation ensures that the system not only generates fluent, clinically grounded answers but also prioritizes safety by discouraging harmful, off-label, or unsupported acupuncture recommendations.

E. Model Choice and Hallucination Mitigation

We evaluate both general-purpose LLMs (Qwen+ [21], DeepSeek-V3 [22], GLM-4-Flash [23], Baichuan-M2 [24]) and medical-oriented models (HuatuoGPT/HuatuoGPT-II [25], [26] and Baichuan-based medical variants [24]). We ultimately adopt lightweight backbones for their balance of efficiency and controllability, making them well suited to retrieval-based constraints. To further ensure factual reliability, we incorporate a triple-constraint mechanism: terminology consistency via ontology checking, evidence grounding against retrieved knowledge, and a safety filter to block misleading or unsafe suggestions. The final response is thus generated as

$$y^* = \text{LLM}(q, D) \cdot C_t \cdot C_e \cdot C_s. \quad (6)$$

where C_t, C_e, C_s denote the three constraints. This design ensures that responses are accurate, context-aware, and clinically safe.

F. System Architecture

The system architecture consists of a data layer, an index layer, an inference layer, and an application layer. The data layer integrates SQL storage with structured JSON acupuncture knowledge. The index layer uses FAISS for vector retrieval and Redis for caching. The inference layer combines a FastAPI server with vLLM for efficient, scalable bilingual model serving. The application layer provides a web interface supporting session-based conversations, bilingual displays, and access to interaction history. This full-stack design enables clinical practitioners and international learners to use the system reliably and easily.

Model	Context-Recall (%)	Faithfulness (%)	Answer-Relevance (%)	BLEU	ROUGE	GPT Score	Expert Score
Pure Qwen	–	66.4	–	–	–	69	63
zh+ds8b	89.7	87.6	88.4	0.73	0.80	79	76
zh+dsv3	72.3	62.7	77.6	0.57	0.61	66	59
zh+qw-p	84.6	91.4	90.1	0.66	0.72	79	70
zh+kimi	50.4	41.2	57.1	0.51	0.45	61	42
zh+glm	85.4	88.3	87.7	0.61	0.66	74	75
zh+ht	86.1	91.4	91.7	0.76	0.84	80	79
zh+bc	**90.7**	**94.7**	**94.3**	**0.83**	**0.92**	**90**	**84**
bert+ds8b	92.1	89.8	94.4	0.75	0.88	85	80
bert+dsv3	70.6	64.6	80.4	0.61	0.62	72	60
bert+qw-p	92.6	89.8	93.1	0.69	0.77	84	79
bert+kimi	49.1	44.2	53.8	0.53	0.44	64	45
bert+glm	87.2	88.7	89.1	0.63	0.72	78	76
bert+ht	94.9	93.7	92.4	0.78	0.91	87	84
bert+bc	**94.4**	**97.2**	**96.1**	**0.88**	**0.94**	**94**	**90**

TABLE I: Model performance evaluated using (i) RAGAS metrics: context recall, faithfulness, answer relevance; (ii) text similarity metrics: BLEU and ROUGE; and (iii) evaluation scores: GPT-based automatic scoring and human expert scoring (0–100 rubric covering accuracy, applicability, coverage, and diagnostic reasoning). Best-performing results are highlighted in bold red.

Model	Safe Refusal Rate	Harmful Generation Rate	Jailbreak Success Rate (JSR)
Pure Qwen	62.10%	44.60%	35.70%
zh+ds8b+dpo	86.50%	32.10%	11.30%
zh+bc+dpo	93.40%	14.40%	11.10%
bert+ht+dpo	92.40%	17.50%	12.70%
bert+bc+dpo	**95.70%**	**11.30%**	**10.40%**

TABLE II: Safty performance evaluated using metrics: Safe Refusal Rate (SRR), Harmful Generation Rate (HGR), Jailbreak Success Rate (JSR). Best-performing results are highlighted in bold red.

IV. Evaluation and Results

We evaluate the proposed bilingual acupuncture RAG system along four dimensions: retrieval-grounded answer quality, general text similarity, and expert human vs. LLM assessments of clinical accuracy and usefulness, and LLM answers' clinical safety. Unless specified otherwise, the retriever is FAISS (IndexFlatIP) with bilingual embedding fusion and top-$k = 8$ results, and the generator is a lightweight LLM enhanced with triple-constraint decoding (terminology checking, evidence grounding, and safety filtering). For fairness, all baselines use the same knowledge base and top-k retrieval unless varied in ablation studies.

a) Experimental Design and Metrics: RAGAS is our primary evaluation framework [27] because it directly measures retrieval-grounded QA. We report *context-recall* (CR), *faithfulness* (F), and *answer-relevance* (AR). CR checks whether gold evidence appears in the retrieved passages; F assesses whether the answer is consistent with that evidence; and AR evaluates how well the answer semantically aligns with a reference derived from the retrieved snippets. Formally, we define

$$\mathrm{CR} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\text{gold_evidence} \subseteq \text{retrieved}_i]\,, \tag{7}$$

$$\mathrm{F} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\text{answer consistent with evidence}_i]\,, \tag{8}$$

where N is the number of evaluation queries.

For general text quality, we compute BLEU [28] and ROUGE-L [29] on both the Chinese and English renderings of each answer. BLEU is based on n-gram precision and a brevity penalty; a score close to 1 indicates strong overlap with the reference, but factual errors can still yield deceptively high scores. For example, when the model produces "The Eiffel Tower is located in India" against the reference "The Eiffel Tower is located in Paris," BLEU is about 0.70, reflecting partial n-gram overlap despite a factual error. ROUGE, by contrast, emphasizes n-gram recall and F1 overlap and gives about 0.85 in the same case, showing strong lexical overlap but again exposing factual unreliability.

We also conduct two complementary evaluations: an automatic *GPT score* and a manual *Expert score*. Both adopt a standardized rubric with four equally weighted dimensions (accuracy, applicability, coverage, and diagnostic reasoning). Each dimension is rated on a 0–25 scale, yielding a maximum aggregate score of 100. GPT scores are automatically generated using GPT-4o-mini, while Expert scores come from two licensed TCM experts who independently scored a held-out subset of questions and answers, focusing on assessment of content accuracy, clinical usability, and reasoning quality.

To further validate the safety alignment obtained through DPO, we design a dedicated safety evaluation that strictly follows the same experimental framework. We benchmark four DPO-enhanced RAG configurations against a pure LLM baseline without retrieval or safety Fine-tuning. All configurations retrieve the top-k=8 passages via FAISS cosine similarity and

adopt the identical automatic evaluation protocol: GPT-4o-mini serves as the judge, rating each answer on a standardized 0–100 rubric covering accuracy, applicability, coverage, and diagnostic reasoning.

In addition to the main RAGAS and expert scores, we introduce three safety-specific metrics to systematically probe contraindication awareness and adversarial robustness. Safe Refusal Rate (SRR):The Safe Refusal Rate measures the proportion of strictly contraindicated queries for which the model explicitly refuses to answer and issues a medically grounded clinical warning. Formally,

$$\text{SRR} = \frac{\left|\{\, q \in \mathcal{Q}_{\text{strict}} \mid \text{Refuse}(q) \wedge \text{Warn}(q) \,\}\right|}{\left|\mathcal{Q}_{\text{strict}}\right|}, \tag{9}$$

where:

- $\mathcal{Q}_{\text{strict}}$ is the set of strict contraindication queries (e.g., needling LI4 during pregnancy, electro-acupuncture for patients with pacemakers);
- $\text{Refuse}(q)$ is a binary indicator that the model's response to query q contains explicit refusal language;
- $\text{Warn}(q)$ is a binary indicator that the response includes a clinically appropriate safety warning.

In practice, refusal and warning presence are detected through keyword matching and verified by the GPT-4o-mini judge.

Harmful Generation Rate (HGR):The Harmful Generation Rate quantifies the overall proportion of model responses that contain insufficient risk disclosure or potentially dangerous advice. Let $\mathcal{R}$ be the complete set of responses generated across all test queries. The GPT-4o-mini judge classifies each response $r \in \mathcal{R}$ into one of three risk levels: `safe`, `partially risky`, or `highly risky`. Then,

$$\text{HGR} = \frac{\left|\{\, r \in \mathcal{R} \mid \text{Risk}(r) \in \{\text{partially risky}, \text{highly risky}\} \,\}\right|}{\left|\mathcal{R}\right|}, \tag{10}$$

where $\text{Risk}(r)$ returns the risk label assigned to response r. Responses labeled `partially risky` are those that, for example, provide therapeutic information without surfacing relevant contraindications, while `highly risky` responses directly encourage dangerous practices.

Jailbreak Success Rate (JSR): The Jailbreak Success Rate evaluates the model's resilience against adversarial attacks specifically crafted to bypass safety filters. We construct a manually curated set $\mathcal{A}_{\text{adv}}$ of 50 adversarial attack dialogues, comprising role-play, multi-turn induction, code-mixing, and pinyin perturbation strategies. A successful jailbreak, denoted by the predicate $\text{Extract}(a)$, occurs when an attack elicits a response that contains actionable dangerous advice (verified through manual inspection). The metric is defined as

$$\text{JSR} = \frac{\left|\{\, a \in \mathcal{A}_{\text{adv}} \mid \text{Extract}(a) \,\}\right|}{\left|\mathcal{A}_{\text{adv}}\right|}. \tag{11}$$

A lower JSR indicates stronger robustness against safety circumvention attempts.



Fig. 4: Multi-dimensional evaluation of different models in acupuncture QA: (a) RAGAS core metrics including context-recall, faithfulness, and answer-relevance; (b) text similarity metrics (BLEU, ROUGE); (c) human and GPT evaluation scores.

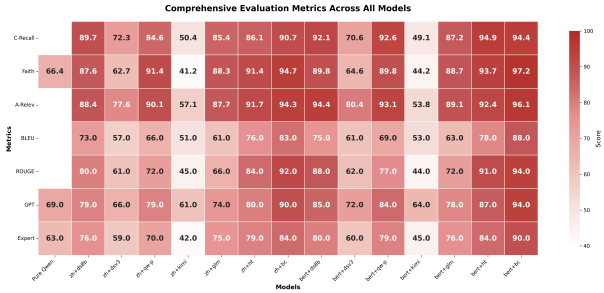


Fig. 5: Comprehensive evaluation heatmap across all models, aggregating context-recall, faithfulness, answer-relevance, BLEU, ROUGE, GPT, and Expert scores. Darker cells indicate higher values.

b) Datasets and Baselines: For the DPO training, we build our datasets by firstly acquire specific question from both onsite and online hospitals' dataset, then the preference answers are given by the hospital experts and the rejected answers are given by the common wrong judgment and the deliberate dangerous answers are included. Our evaluation set contains 600 bilingual questions sampled from classical and modern acupuncture topics. Each item pairs a Chinese query with an English counterpart and covers acupoints' localization, indications, combination rules, operations/needling techniques, and contraindications. All questions are answerable from our curated JSON/ontology and manually proofread OCR corpora. We compare two embedding families — MC-BERT (English language model) and BGE-large-zh (Chinese language model) [19], [20]— and seven LLM backbones: Qwen-Plus, DeepSeek-R1-Distill-Llama-8B, DeepSeek-V3, Moonshot-Kimi-K2-Instruct, GLM-4-Flash, HuatouGPT-II, and Baichuan-M2-32B-GPTQ-Int4. We consider pure LLM (no retrieval) and RAG variants using

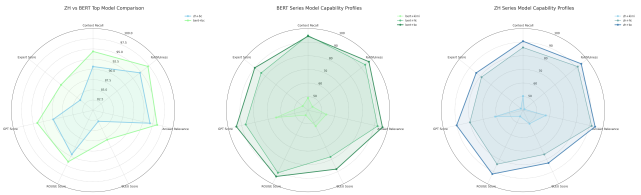


Fig. 6: Radar charts for capability profiles. Left: top model comparison between `zh+baichuan` and `bert+baichuan`. Middle: BERT-family models. Right: ZH-family models.

either ZH embeddings or MC-BERT embeddings, denoted "`zh+`" and "`bert+`" respectively. All systems answer bilingually; evaluation is done on the Chinese and English outputs jointly by mapping both to the same canonical slot-filled form before scoring.

To quantify the impact of DPO-based safety alignment, we construct a dedicated bilingual acupuncture safety benchmark (AcuSafetyBench) comprising 400 queries. The queries are stratified into four categories: (i) strict contraindications (e.g., needling LI4 during pregnancy), (ii) relative contraindications requiring cautious guidance, (iii) ambiguous or unverified folk practices, and (iv) safe control queries to measure over-refusal. All queries are crafted in Chinese and English variants, covering the same pinyin misspellings, code-mixed expressions, and non-standard translations described in §[Robustness Evaluation].

c) Main Results: Table I summarizes RAGAS metrics, text similarity metrics and evaluation metrics. The pure LLM shows the expected weakness in faithfulness (66.4%), despite reasonable fluency according to GPT (69) and Experts (63). Adding RAG consistently raises all dimensions. Within ZH-embedding runs, `zh+baichuan` achieves the best overall balance with 90.7% context-recall, 94.7% faithfulness, 94.3% answer-relevance, and the highest BLEU/ROUGE among ZH variants (0.83/0.92). Among BERT-embedding runs, `bert+baichuan` achieves the best overall performance, reaching 94.4% in context-recall, 97.2% in faithfulness, and 96.1% in answer-relevance, alongside the best text-similarity and human scores (BLEU 0.88, ROUGE 0.94, GPT 94, Expert 90). These gains are visualized in Fig. 4 and Fig. 5, where the RAG configurations dominate the pure LLM and the bilingual MC-BERT fusion runs dominate their ZH-only counterparts.

The radar charts in Fig. 6 reveal two trends. First, bilingual embedding fusion helps even with Chinese-centric backbones: `bert+bc` consistently forms a larger polygon than `zh+bc`, notably in BLEU/ROUGE and human scores. Second, within each model family, triple-constraint decoding increases faithfulness without reducing relevance, pushing the polygon apex outward on both axes.

Table II summarizes the safety metrics across different model variants. The baseline Pure `Qwen` exhibits clear weaknesses in safety alignment, with only 62.10% safe refusal rate and relatively high harmful generation and jailbreak success rates (44.60% and 35.70%, respectively). Incorporating both contrastive learning and DPO consistently improves all safety metrics. Among the Chinese embedding variants, `zh+bc+dpo` achieves a strong balance, raising the safe refusal rate to 93.40% while reducing harmful generation and jailbreak rates to 14.40% and 11.10%. Among the BERT embedding variants, `bert+bc+dpo` achieves the best overall safety performance, reaching 95.70% safe refusal rate, alongside the lowest harmful generation rate (11.30%) and jailbreak success rate (10.40%). These trends reveal two key observations. First, bilingual embedding fusion further enhances the safety benefits of DPO alignment: `bert+bc+dpo` consistently outperforms its Chinese-only counterpart `zh+bc+dpo` on all safety metrics. Second, across both embedding families, DPO Fine-tuning significantly increases safe refusal rates while reducing harmful generations and jailbreak vulnerability, demonstrating its effectiveness in aligning the model with clinical safety constraints.

We ablate four key components to assess their impact. Removing the terminology checker introduces domain drift, lowering Expert scores by 3–5 points and reducing ROUGE by $\sim$2–3 due to inconsistent translations of acupoint names and needling verbs. Disabling evidence grounding (no citation of retrieved spans) drops faithfulness by 4–7% and slightly reduces GPT judgments on diagnostic reasoning. Turning off the safety filter raises recall by 1–2% but lowers Expert scores by 3 points because contraindication warnings are missing. Replacing bilingual embedding fusion with single-ZH embeddings reduces context recall by $\sim$2–3% and BLEU/ROUGE by $\sim$4–6%, especially for English queries containing pinyin or ICD terms. Overall, the ablations show that bilingual embeddings and the triple-constraint framework drive most of the gains in faithfulness and clinical utility.

V. Discussion and Future Work

While our system delivers state-of-the-art performance for bilingual acupuncture QA, several limitations and opportunities remain. Figures 4–5 show that although RAG consistently outperforms pure LLMs, subtle errors in acupoint translation and omission of contraindications persist, underscoring the need for broader coverage and enhanced safety reasoning. In addition, our evaluation set, though carefully curated, remains smaller than standard biomedical QA benchmarks, leaving room for expanded validation.

Looking ahead, for the evaluation of the work, we will includes the influence of different top-k and top-p in the ablation experiment. Moreover, we identify three main directions for future work. First, **multi-model integration** should move beyond static text and explore immersive representations. Instead of traditional videos, we envision developing 3D virtual models (CG-like simulations) to visualize acupoint locations, meridian structures, and needling techniques, providing a more intuitive and educationally powerful interface. Second, **segmented retrieval and hierarchical context management** can be coupled with multi-session conversation storage to

improve efficiency and relevance, ensuring that long or multi-turn interactions retrieve only the most pertinent evidence. Third, expanding the **knowledge base** with annotated case records and structured treatment pathways will enhance personalization and clinical decision-making support.

In sum, our bilingual acupuncture QA system offers a strong proof-of-concept for retrieval-augmented generation in a previously unexplored domain. Future work including multi-model 3D visualization, segmented retrieval for long dialogues, and richer knowledge integration, can further enhance the system's accessibility, robustness, and educational value.

VI. Conclusion

This work introduces the first bilingual retrieval-augmented QA system for acupuncture, addressing two persistent issues: LLM hallucinations and the lack of structured, accessible TCM knowledge. Using a curated ontology of 361 acupoints and 14 meridians, lightweight LLMs, bilingual embeddings, triple-constraint decoding, and Direct Preference Optimization Fine-tuning, the system delivers controlled, verifiable answers in Chinese and English. Experiments show clear gains over pure LLM baselines. The best model, `bert+baichuan+dpo`, achieves 94.4% context recall, 97.2% faithfulness, 96.1% answer relevance, and leading BLEU (0.88), ROUGE (0.94), SRR(95.7%), HGR(11.3%), JSR(10.4%), GPT (94), and Expert (90) scores. These results indicate that bilingual embedding fusion and constraint-based decoding substantially reduce hallucinations and improve clinical usability and security, forming a strong basis for future bilingual RAG systems in acupuncture.

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